Foster + Partners designs exoskeletons for stepped-back skyscrapers in Manila

British studio Foster + Partners has begun construction on two stepped-back towers for the BDO Unibank Inc campus in the Philippines, which will have exoskeletons to support solar shading and achieve structural stability. The studio is transforming five urban plots in Manila to create the campus, with the two towers – which are connected by a covered walkway – rising above open ground floors. Foster + Partners looked to vernacular architecture for the buildings, which will step back at regular intervals. At each step back, there are triple-storey amenity spaces as well as outdoor terraces. The towers have a stepped-back designThe two towers at BDO Unibank Inc campus will each have an exoskeleton, designed to support woven metal-mesh infill screens. These take visual cues from traditional weaving patterns and will shade the areas that receive the most sunlight. The exoskeletons will also be seismic-resistant, which the studio said is needed in the region. "The structure of the buildings is intentionally placed on the outside of their envelopes to actively shade the facades, while achieving an efficient structural design in a seismic zone," Foster + Partners senior partner Roland Schnizer said. "The exoskeletons create column-free spaces internally and support external solar shading screens." Artworks will be on display inside the campusAccording to Foster + Partners, the structural system reduces the concrete content of the buildings by more than 65,000 tonnes. Visitors to the campus, which will contain offices, a business centre, a public museum and a canteen, will enter on the ground floor before reaching an elevated reception. On the open ground plane, the campus will feature artworks and plants to "enhance the public realm". An urban farm on the roof of BDO Unibank Inc campus will be used to grow food for the staff canteen, while more than 70 per cent of the campus' potable water will be recycled and reused. An exoskeleton will be used for seismic resistanceThe studio is aiming to achieve Green Mark Super Low Energy, which is the highest certification issued by the green building rating system Green Mark Building. It is awarded to projects that achieve at least 60 per cent energy savings when compared to 2005 building codes. If achieved, it would be the first building in the Philippines with the rating. "The new BDO campus offers a new model for the next generation of highly flexible and climatically responsive workplaces in the Philippines," Foster + Partners head of studio Luke Fox said. "Our holistic approach encompasses every element of the design – from the structural and environmental engineering to the landscaping and interiors – allowing us to create something completely bespoke and driven by extensive environmental analysis." Foster + Partners also recently unveiled a pair of Dubai skyscrapers linked by a swimming pool and added a barrel-vaulted structure to a Spanish winery.







